

[5869] - 416

S.E. (Computer/Information Technology)
ENGINEERING MATHEMATICS - III
(2015 Pattern) (Semester - II) (207003)

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figures to the right indicate full marks.
- 3) Your answers will be valued as a whole.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Solve any two

[8]

i) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = \frac{1}{1+e^{3x}}$

ii) $\frac{d^2y}{dx^2} + 4y = e^x + x^2$

iii) Solve by Method of Variation of parameters $(D^2 - 2D)y = e^x \sin x$

b) Find the fourier cosine integral representation of $f(x) = \begin{cases} x^2 & : 0 < x < a \\ 0 & : x > a \end{cases}$

[4]

OR

Q2) a) Solve : $\frac{dx}{2x} = \frac{dy}{-y} = \frac{dz}{4xy^2 - 2z}$

[4]

b) Solve any one :

[4]

i) Find Z-transform of $f(k) = k^2 e^{-ak}$; $k \geq 0$.

ii) Find $Z^{-1} \left[\frac{3Z^2 + 2Z}{Z^2 - 3Z + 2} \right]$; $1 < |Z| < 2$.

c) Solve the following difference equation $6f(k+2) - f(k+1) - f(k) = 0$,
 $f(0) = 0, f(1) = 1$.

[4]

P.T.O.

Q3) a) Find first four moments about mean if the first four moments about 10 are 3, 40, 100 and 400. Also find β_1 & β_2 . [4]

b) Fit least squares straight line to the following data and use it to find the value of y at $x = 6$ [4]

x	1	2	3	4	5
y	0	0	2	6	12

c) A machine produces 2% defective bulbs. Bulbs are sold in a box of 100 bulbs, with a guarantee that no more than 3 bulbs are defective. Use Poisson distribution to determine the probability that a box will fail to meet the guarantee. [4]

OR

Q4) a) A problem of statics is given to 5 students A, B, C, D and E. Their chances of solving the problem are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ and $\frac{1}{6}$. What is the probability that problem is solved. [4]

b) In a certain factory turning out razor blades there is a small chance of $\frac{1}{500}$ for any blade to be defective. The blades are supplied in a packet of 10. Calculate approximate number of packets containing no defective and 2 defective blades in a consignment of 10,000 packets. [4]

c) Find correlation coefficient between X and Y , given that, $n = 25$, $\Sigma x = 75$, $\Sigma y = 100$, $\Sigma x^2 = 250$, $\Sigma y^2 = 500$, $\Sigma xy = 325$. [4]

Q5) a) If the directional derivative of $\phi = axy + byz + czx$ at $(1, 1, 1)$ has maximum magnitude 4 in a direction parallel to x -axis, find the values of a, b, c . [4]

b) Find $\text{Cur} | \text{Cur} | \bar{F}$ at the point $(0, 1, 2)$, where $\bar{F} = x^2 y \bar{i} + xyz \bar{j} + z^2 y \bar{k}$. [4]

c) Evaluate $\int_c \bar{F} \cdot d\bar{r}$ for $\bar{F} = 3x^2 \bar{i} + (2xz - y) \bar{j} + z \bar{k}$ along the straight line joining $(0, 0, 0)$ and $(2, 1, 3)$. [5]

OR

Q6) a) Find the directional derivative of $\phi = x^2 - y^2 - 2z^2$ at the point $P(2, -1, 3)$ in the direction PQ where Q is $(5, 6, 4)$. [4]

b) Show that $\vec{F} = \frac{xi + yj}{x^2 + y^2}$ is solenoidal and irrotational. [4]

c) Find the work done in moving a particle once round the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1, z = 0$ under the field of force given by $\vec{F} = (2x - y + z)\vec{i} + (x + y - z^2)\vec{j} + (3x - 2y + 4z)\vec{k}$ [5]

Q7) a) Determine the analytic function $f(z) = u + iv$ if $u = x^3 - 3xy^2$. [4]

b) Find the bilinear transformation, which sends the points $1, i, -1$ from z - plane into the points $i, 0, -i$ of the w - plane. [4]

c) Evaluate $\oint_c \frac{2z^2 + z + 5}{(z - \frac{3}{2})^2} dz$, where 'c' is the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$. [5]

OR

Q8) a) Find the condition under which $u = ax^3 + bx^2y + cxy^2 + dy^3$ is harmonic. [4]

b) Find the bilinear transformation, which maps the points $z = -1, 0, 1$ on to the points $w = 0, i, 3i$. [4]

c) Evaluate $\oint_c \frac{(z^2 + \cos^2 z)}{(z - \frac{\pi}{4})^3} dz$, where c is $|z| = 1$. [5]

